

Practical 03

Aim: To study and execute DCL and TCL Commands.

Theory:

Data Control Language (DCL):

DCL (Data Control Language) includes commands such as **GRANT** and **REVOKE** which mainly deal with the **rights, permissions**, and other controls of the **database system**. These commands are used to **control access** to data in the database by **granting** or **revoking permissions**.

1. GRANT:

Assigns new privileges to a user account, allowing access to specific database objects, actions, or functions.

Syntax

GRANT privilege_type [(column_list)] ON [object_type] object_name TO user [WITH GRANT OPTION];

2. REVOKE:

Removes previously granted privileges from a user account, taking away their access to certain database objects or actions.

Syntax

REVOKE [GRANT OPTION FOR] privilege_type [(column_list)] ON [object_type] object_name FROM user [CASCADE];

Write down queries to be executed in practical slot related to DCL Command.

Transaction Control Language (TCL)

Transactions group a set of tasks into a **single execution unit**. Each transaction begins with a specific task and ends when all the tasks in the group are successfully completed. If any of the **tasks fail**, the transaction fails. Therefore, a transaction has only two results: **success** or **failure**.

1. COMMIT:

Saves all changes made during the transaction.

Syntax

COMMIT;

2. ROLLBACK

Undoes all changes made during the transaction.

Syntax

ROLLBACK;

3. SAVEPOINT:

Creates a savepoint within the current transaction.

Syntax

SAVEPOINT savepoint_name;

Write down queries to be executed in practical slot related to TCL Command.

RESULT:

The DCL and TCL Commands are executed successfully.